

Market Status Framework Design Notes

Core Principle

Do not classify an instrument as stressed simply because it is up or down a lot. Measure whether it is behaving unusually relative to its own history and whether the move conveys information about the broader market regime.

Three Layers

- 1. Observation:** collect normalized facts (returns, ATR/range, moving averages, volume, percentile moves, key-level breaks, persistence).
- 2. Instrument Status:** classify Quiet, Active, Stressed, or Extreme using a scoring model.
- 3. Market Role:** classify the instrument as Leading, Confirming, Diverging, or Isolated.

Scoring Model

Move Magnitude (0–3): normalize return by expected daily move (ATR or realized volatility).

Range Expansion (0–2): compare true range to ATR20.

Technical Displacement (0–2): reward meaningful support/resistance or moving-average breaks.

Volume Confirmation (0–2): compare against normal volume (or same time-of-day for futures).

Persistence (0–2): distinguish accepted repricing from fade.

Suggested Status Mapping

0–1 = Quiet

2–3 = Active

4–6 = Stressed

7–9 = Extreme

Market Roles

Leading: moves before the broader market confirms.

Confirming: follows established leadership.

Diverging: contradicts the expected regime (often very informative).

Isolated: company- or sector-specific without confirmation.

SOXS Example

Do not classify SOXS using raw percentage change. Instead evaluate:

- Volatility-adjusted move percentile
- Whether SMH/SOX broke important levels
- Relative volume
- Excess inverse move versus expected -3x response
- Persistence of gains.

Cross-Asset Regime

Separate instrument status from regime status. Example:

Semiconductor Stress → Leaders: DRAM, Kioxia, SOXS; Confirmations: SMH, NQ, Nikkei;

Non-confirmations: VIX, credit spreads; Regime: Sector Stress.

As volatility broadens later, the regime can evolve into Broadening Risk-Off.

Recommended Data Model

Intensity: Quiet → Active → Stressed → Extreme
Direction: Risk-On / Risk-Off / Inflationary / Deflationary
Role: Leading / Confirming / Diverging / Isolated
Confidence: Low / Medium / High

Example YAML

```
ticker: SOXS
status: stressed
direction: risk_off
role: leading
confidence: high
reasons:
- 97th percentile overnight move
- 1.8 ATR range expansion
- broke 20-day high
- volume 2.1x normal
```

Key Takeaway

The most useful dashboards distinguish facts from interpretation. First compute objective measures, then classify intensity, role, and regime. A 'Quiet but Diverging' VIX can be more informative than a simply 'Stressed' VIX.